Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. COMPUTER SCIENCE AND ENGINEERING

MAKEUP EXAMINATIONS, DEC 2019

SUBJECT: SOFT COMPUTING PARADIGMS (CSE 4031)

REVISED CREDIT SYSTEM
(28-12-2019)

Time: 3 Hours MAX. MARKS: 50

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Give the McCulloh's – Pitt's definition of neural network. Explain the properties and capabilities of a neural network? **5M**

1B. Explain the working of Reinforcement Learning and Unsupervised Learning model. **5M**

2A. Prove that for any two linearly separable classes there is a solution weight vector W0 after at most n_{max} iterations given by **5M**

$$n_{max} = \frac{\beta \|w_0\|^2}{\alpha^2}$$

where α and β are positive constants.

2B. With the help of signal flow graph, derive the weight updating equation of output neuron connected to hidden neuron j. **5M**

3A. Use the back propagation algorithm for computing a set of synaptic weights and bias levels for a neural network to solve the XOR problem. Assume the use of a logistic function for the non-linearity. **5M**

3B. Explain the energy minimization of a discrete Hopfield network. Also explain with example spurious state. **5M**

4A. Explain with a neat diagram learning vector quantization. **4M**

4B. Consider two fuzzy sets given by, **4M**

$A = \left\{\frac{1}{low} + \frac{0.2}{medium} + \frac{0.5}{high}\right\}$ $B = \left\{\frac{0.9}{positive} + \frac{0.4}{zero} + \frac{0.9}{negative}\right\}$

a) Find the fuzzy relation for the Cartesian product of A and B i.e., R = A x B

b) Introduce a fuzzy set C given by,

$$C = \left\{\frac{0.1}{low} + \frac{0.2}{medium} + \frac{0.7}{high}\right\}$$

Find the relation between C and B using Cartesian product i.e., find S = C x B

c) Find C o R using max - min composition.
d) Find C o S using max - min composition.

4C. Explain with an example intuition and inference methods for assigning membership values. **2M**

5A. Explain three main operators in Genetic algorithm and its effects. **5M**

5B. What is the need for defuzzification process? How lambda-cuts for fuzzy sets and fuzzy relations can be carried out. **5M**